PH4102: Introductory Astrophysics

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White Dwarfs

- Typical densities of white dwarfs $\rho \sim 10^9$ kg/m^3

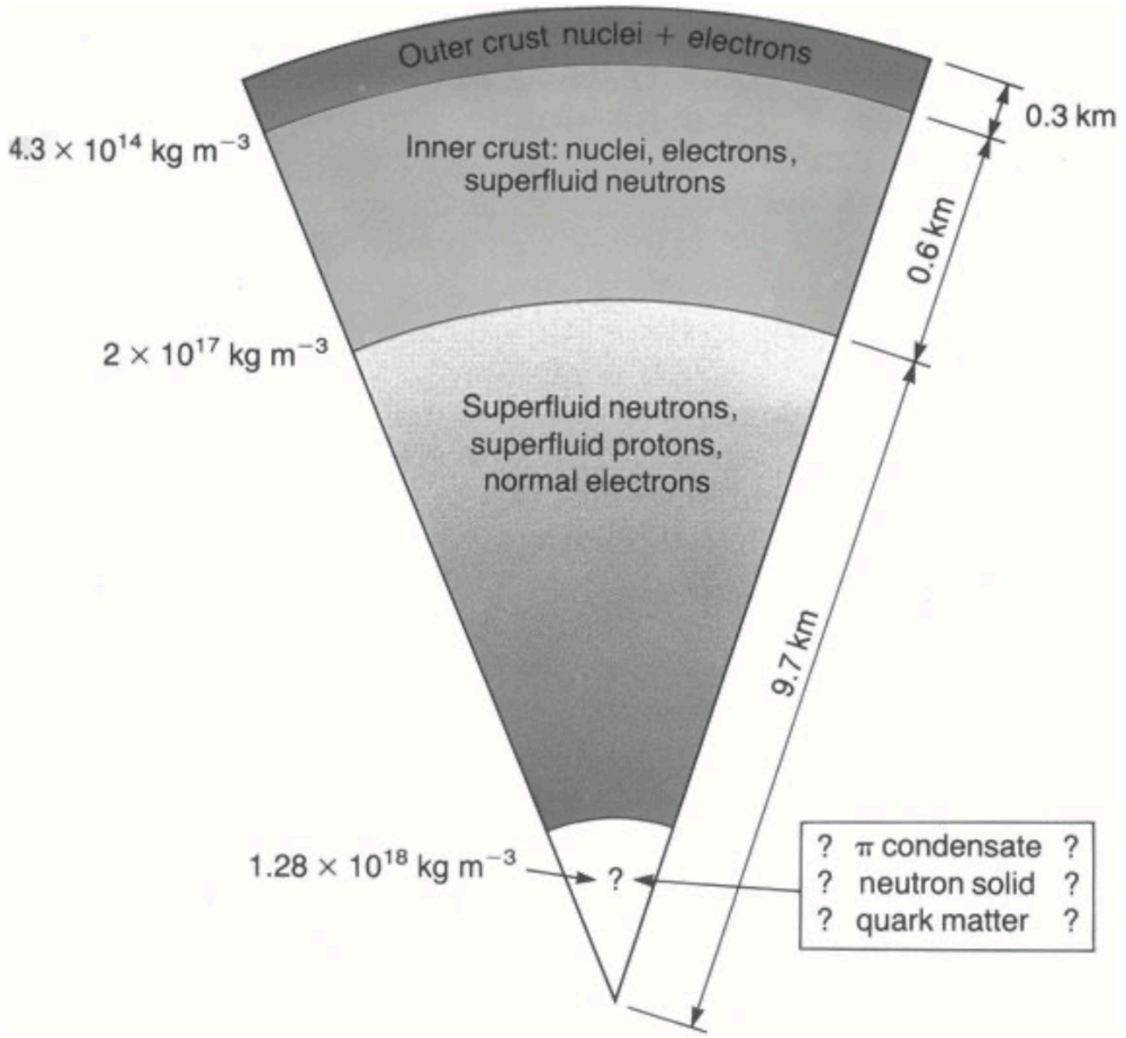
- The main uncertainty is the chemical composition of the star. Spectroscopic observations of their surface properties show that most white dwarfs have lost their hydrogen envelopes. Nuclear burning of massive stars can result in the formation of a helium core -> carbon–oxygen core -> iron core. The fate of these stars therefore depends upon whether there is sufficient mass loss for them to end up as white dwarfs or whether they undergo catastrophic collapse to neutron stars or black holes.

- The thermal energy of the star is derived from the internal energy with which the star was endowed when it was formed. The cooling time for white dwarfs is about $10^9 - 10^{10}$ years, very much longer than the thermal cooling time-scale for a star like the Sun, because of their smaller size (hence, smaller surface area). For the white dwarf stars in star clusters, the ages of the clusters are of the same order as the cooling lifetimes of the white dwarfs. Because of their high surface temperatures and small diameters, the white dwarfs lie below the main sequence on the H-R diagram

Neutron stars

- The interiors of neutron stars consist of zones of increasing density until the material attains nuclear densities in bulk. With increasing density, the degenerate electron gas becomes relativistic and, when the total energy of the electron exceeds the mass difference between the neutron and the proton, $E = \gamma m_e c^2 \geq (m_n - m_p)c^2 = 1.29$ MeV the inverse β decay process $p + e^- \rightarrow n + \nu_e$ can convert protons into neutrons.
- In a non-degenerate electron gas, the neutrons would decay into protons and electrons with a mean lifetime of 14.8 minutes, corresponding to a half-life of 10.2 minutes, but this is not possible if the electron gas is degenerate as there are no available states for the ejected electron to occupy.
- This stabilisation takes place when the Fermi energy of the degenerate electron gas is greater than the kinetic energy of the emitted electrons. For a hydrogen plasma, the critical density at which stabilisation takes place can be found as follows: The total energy of the electron must be $E \geq E_{tot} = (m_n - m_p)c^2 = 1.29$ MeV. The critical Fermi momentum p_F can be found from $E_{tot} = \sqrt{p_F^2 c^2 + m_e^2 c^2}$ and the corresponding critical electron number density can be found to be $n_e = (8\pi/3h^3)p_F^3$. From this, critical mass density can be found, which comes out to be $\rho = 1.2 \times 10^{10} kg/m^3$.
- Above this density, β decay process doesn't take place and neutrons are abundantly form. This process is often referred to as *neutronisation*.

- For heavier nuclei, especially for the nuclei of the conventional stable elements such as carbon, oxygen and iron, the energies needed to achieve this $p \rightarrow n$ transition are greater than in the case of isolated, single protons because the neutrons are degenerate within the nuclei and therefore the electron must be sufficiently energetic to exceed the Fermi energy within the nucleus.
- However, if the nuclei become too neutron-rich, they begin to break up and an equilibrium state is set up consisting of neutron-rich nuclei, a free neutron gas and a degenerate relativistic electron gas. This process of releasing neutrons from the neutron-rich nuclei is referred to as *neutron drip* and sets in at a density of about $4 \times 10^{14} kg/m^3$.
- These processes result in profound changes in the equation of state such that stable stars cannot form until much higher central densities are attained, $\sim 10^{17} kg/m^3$, at which the neutron-drip process has converted almost all of the matter into neutrons. The degeneracy pressure of the neutron gas prevents collapse under gravity and results in the formation of a *neutron star*.
- The underlying physics is the same as for white dwarfs, the difference being that the neutrons are about 2000 times more massive than the electrons and consequently, degeneracy sets in at a correspondingly higher density. In addition, a general relativistic treatment is needed to determine the structures of the most massive neutron stars.
- Detailed computer modelling of neutron stars, along with a very general argument involving the general theory of relativity, shows that the maximum mass possible for a neutron star cannot exceed about 2.2 $M_\odot$ if it is static and 2.9 $M_\odot$ if it is rotating rapidly.



A representative model showing the internal structure of a 1.4 $M_\odot$ neutron star.

i) The outer crust, $10^9 \leq \rho \leq 4.3 \times 10^{14} kg/m^3$, consists of a solid region composed of matter similar to that found in white dwarfs

ii) The inner crust, $4.3 \times 10^{14} \leq \rho \leq 2 \times 10^{17} kg/m^3$, consists of a lattice of neutron-rich nuclei together with free degenerate neutrons and a degenerate relativistic electron gas

iii) The neutron liquid, $\rho \sim 2 \times 10^{17} kg/m^3$, consists mainly of neutrons with a small concentration of protons and electrons

iv) a *core region* of very high density $\rho \geq 3 \times 10^{18} kg/m^3$ may or may not exist. The existence of this phase depends upon the behaviour of matter in bulk at very high energies and densities. It is not clear if there is a phase transition to a neutron solid or to quark matter or to some other phase of matter quite distinct from the neutron liquid. Many of the models of stable neutron stars do not possess this core region but it is certainly not excluded that quite exotic forms of matter could exist in the centres of massive neutron stars.

Observation of Pulsars

- Radio pulsars were discovered first by Hewish and Bell in 1967 (Hewish *et al.*, 1968) and the source PSR 1919+21 (PSR - pulsating source of radio followed by RA in hhmm and DEC) was the first *pulsar* to be identified in November 1967 and over the next few months three further examples were discovered with pulse periods in the range 0.25 to almost 3 s.

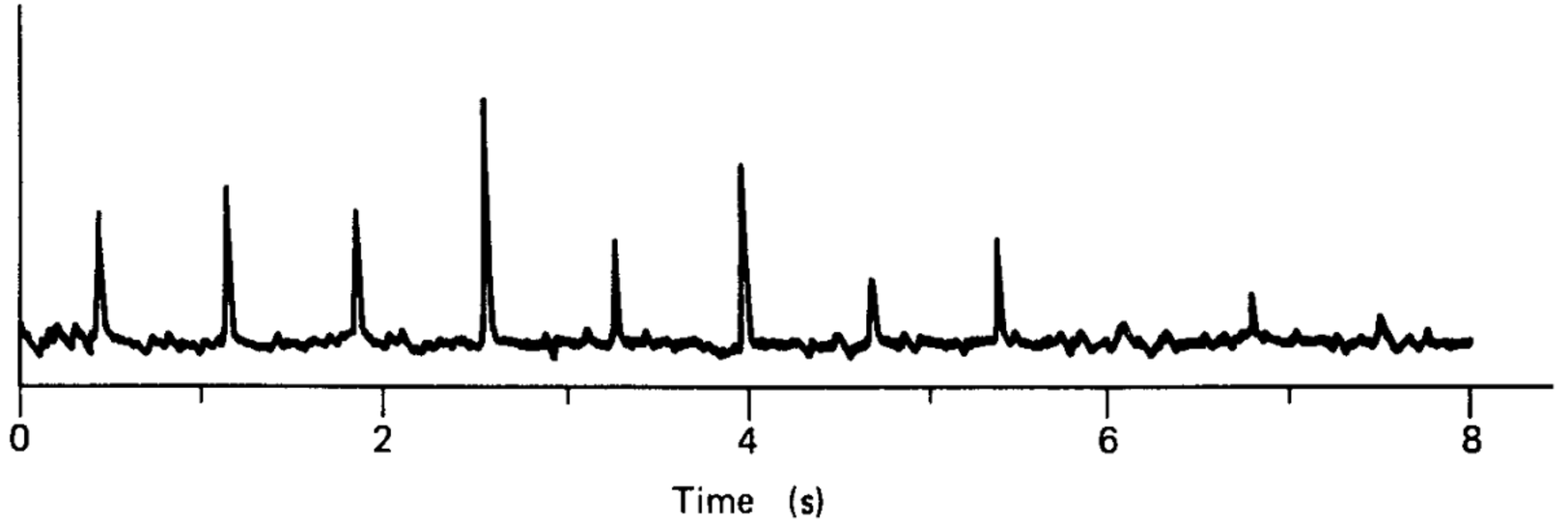


Fig. 5.3 Radio signals from the pulsar PSR 0329 + 54, which has a period of 0.714 s. Note that different pulses are not identical and some pulses are even missing.

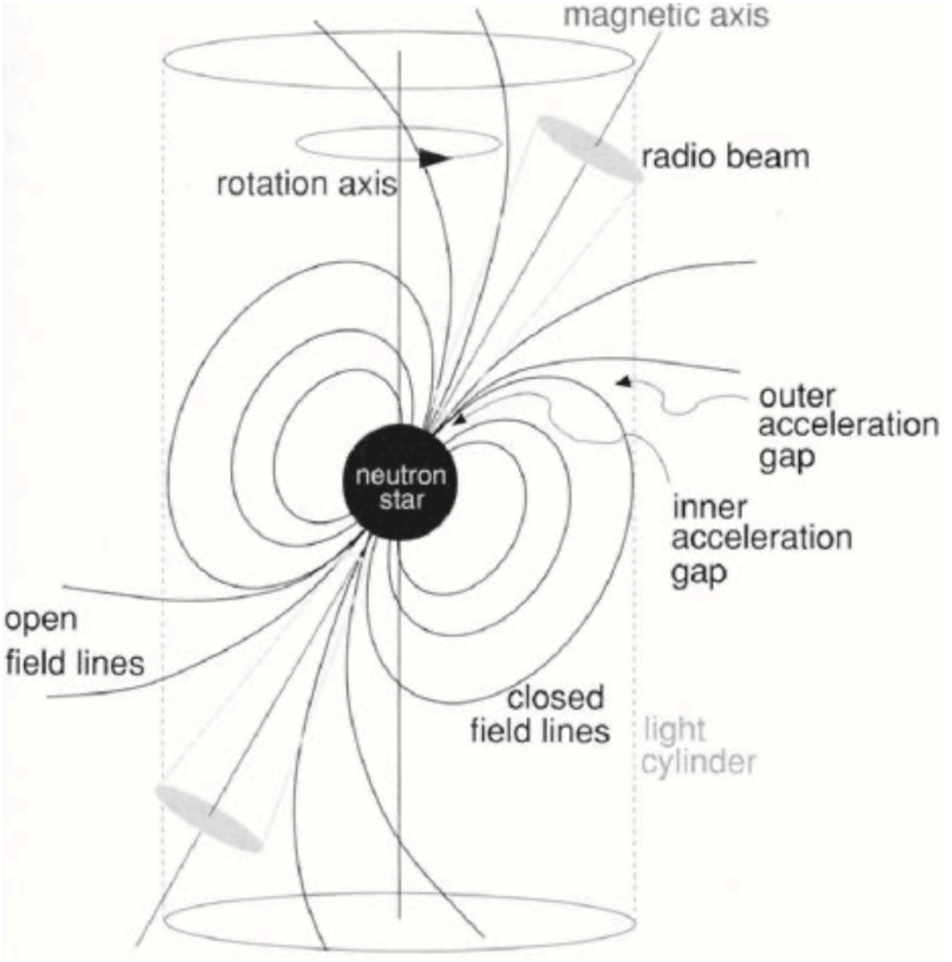
- With improvement of technology, discovery of mili-second pulsars (pulse period ~ 10 millisecond or less) has become possible later.

Soon after the discovery, Gold (1968) identified pulsars as rotating neutron stars. The pulse period must be due to some physical mechanism like rotation or oscillation. Theoretical estimates of oscillation periods of white dwarfs or neutron stars show that they do not match the observed pulsar periods (oscillation periods of normal stars are much longer). If the pulsar period has to be identified with the rotation period of some object, one has to make sure that the centrifugal force is not stronger than gravity, i.e.,

$$\Omega^2 r < GM/r^2 \rightarrow \Omega < (G\rho)^{1/2}$$

A rotation period of 1 s demands that the rotating object should have a density higher than $10^{11} kg/m^3$ if it is not to be disrupted by the centrifugal force. The pulsars with shortest periods could not be rotating white dwarfs (less density than this!). The only possibility is that the pulsars are rotating neutron stars.

How are the pulses produced?



A schematic model of a pulsar as a magnetised rotating neutron star in which the magnetic and rotation axes are misaligned. The radio pulses are assumed to be due to beams of radio emission from the poles of the magnetic field distribution and are associated with the passage of the beam across the line of sight to the observer (Lorimer and Kramer, 2005). Typical neutron star parameters are $M \approx 1.4\ M_{\odot}$, radius $\approx$ 10 km, magnetic flux density 10^5—10^9 T.